

Joy Shaha

Senior Software Engineer

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LinkedIn | GitHub

SUMMARY

Software Engineer with 7+ years of experience designing and delivering scalable backend systems, cloud-native applications, and AI/ML-powered fintech solutions. Strong expertise in system architecture, distributed systems, and full-stack development using Python and modern JavaScript frameworks. Proven ability to lead technical design, optimize system performance, and deliver production-grade platforms in healthcare, fintech, and enterprise domains.

EXPERIENCE

Senior Software Engineer

October, 2021 – Present

Techcare Inc.

Dhaka, Bangladesh

- Lead backend architecture and development for a cloud-native mental health platform serving production users globally.
- Designed scalable system architecture for therapy booking, user management, payments, and clinical workflows.
- Integrated payment gateways, SMTP/email services, and third-party APIs to support transactional workflows.
- Collaborate with cross-functional teams to define technical roadmap and system design decisions.
- Improved system reliability and maintainability through modular backend design and API standardization.

IoT Trainee | Software Engineer

October, 2017 – October, 2021

Datasoft Systems Bangladesh Ltd.

Dhaka, Bangladesh

- Built AI-powered Anti-Money Laundering (AML) system using machine learning models on banking transaction data.
- Designed and implemented entity resolution systems to detect duplicate identities and fraudulent activity patterns.
- Developed scalable REST APIs and backend services for fintech platform Remit365 handling remittance workflows.
- Implemented role-based access control, reporting pipelines, and transaction processing systems connecting to the banking system.
- Worked on both backend and frontend using Django, JavaScript, delivering end-to-end enterprise features.

TECHNICAL SKILLS

Backend: Python (Django, FastAPI, Flask), Node.js (Express)

Frontend: React, Redux, RTK Query, JavaScript, HTML, CSS (Tailwindcss, bootstrap)

Databases: PostgreSQL, MySQL, MongoDB, Oracle

Cloud: AWS (Lambda, EC2, S3, SQS, SNS, API Gateway), GCP(App Engine), Azure

DevOps: Docker, Nginx(L4/L7), Kubernetes (K3s), CI/CD (GitHub Actions, CodePipeline)

Messaging/Streaming/Async: Celery, Redis, RabbitMQ, Kafka

Architecture: Microservices, Monolith-to-microservice migration, Event-driven systems

AI Tools: Claude(Claude Code), Chatgpt(Codex), Cursor, OpenClaw

SYSTEM DESIGN & ARCHITECTURE

- Designed scalable backend systems using microservices and modular monolith(mono-repo or poly-repo) architecture.
- Built asynchronous task processing pipelines using Celery and Redis.
- Implemented event-driven architecture using AWS services (SQS, SNS, EventBridge).
- Designed containerized deployment pipelines using Docker and CI/CD automation.
- Worked with Nginx, Gunicorn, and Uvicorn for production-grade service deployment.

PRODUCTS & PROJECTS

DiveThru

Edmonton, Alberta, Canada

Health Tech Platform

October, 2021 – Present

Technologies: FastAPI, React, MySQL, MongoDB, AWS, AWS Powertool, Docker

- Designed and developed core healthcare workflows including appointment scheduling, therapist matching, intake questionnaires, treatment notes, and client management.
- Built secure, scalable cloud-native services using FastAPI, AWS Lambda, API Gateway, S3, CloudFront, EventBridge, SNS, and SQS, implementing authentication, authorization, and role-based access control.
- Developed payment and billing integrations supporting therapy sessions, subscriptions, and insurance workflows while collaborating with cross-functional stakeholders to deliver HIPAA-conscious solutions.

Remittance Management System (Remit365)

Dhaka, Bangladesh

FinTech Remittance Platform

March, 2019 – 2021

Technologies: Django(DRF), React, PostgreSQL, Redis, Celery, On-Premise Infrastructure

- Developed core remittance processing workflows and integrated banking partners to automate transaction disbursement and reconciliation.
- Designed and implemented REST APIs, beneficiary management, reporting services, and role-based access control for secure financial operations.

Anti-Money Laundering (AML)

Dhaka, Bangladesh

FinTech Compliance Platform

March, 2018 – 2019

Technologies: Flask, .Net, MSSQL, Azure Machine Learning Studio

- Developed machine learning-driven Anti-Money Laundering (AML) solutions using large-scale banking transaction data to detect suspicious activities and support regulatory compliance. Implemented risk-scoring, entity resolution, and record-matching algorithms to identify fraudulent transactions, linked customer identities, and improve compliance reporting accuracy.

PUBLICATIONS

- Intelligent Power Management System Using ANN - IEEE
- Arduino-based ANN Power Management System - STM Journals

ACHIEVEMENTS

- ACMP-4.0, Institute of Business Administration (IBA), University of Dhaka

EDUCATION

B.Sc in Electrical and Electronic Engineering

2012 – 2016

American International University-Bangladesh

Dhaka, Bangladesh

CGPA: 3.85/4.00